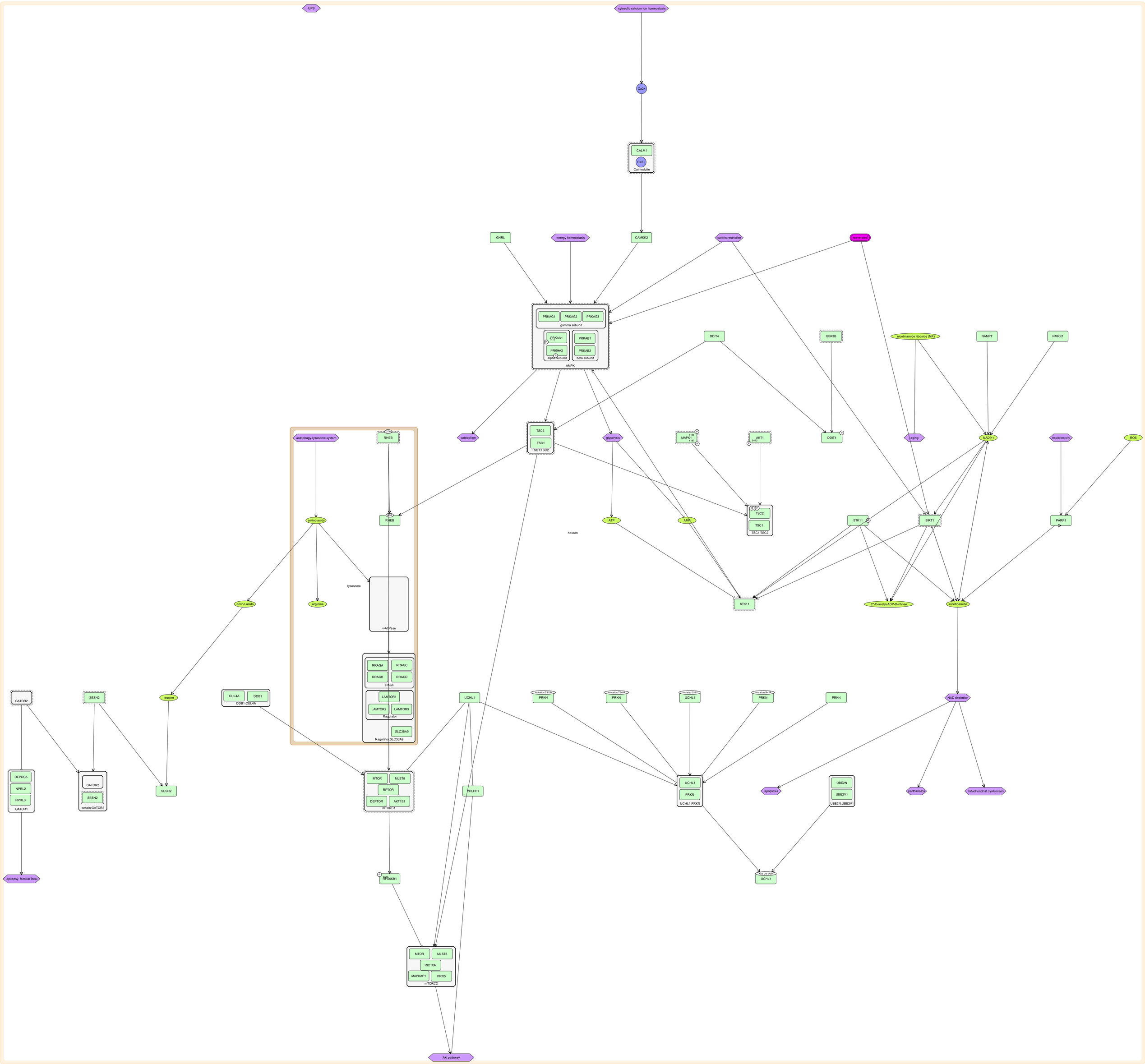
aging
nicotinamide riboside (NR)
NMRK1
NAD(+)
resveratrol
NAD(+)
STK11
Ac
2"-O-acetyl-ADP-D-ribose
STK11
SIRT1
NAMPT
caloric restriction
PARP1
ROS
excitotoxicity
nicotinamide
PARP1
NAD depletion
parthanatos
apoptosis
mitochondrial dysfunction
catabolism
ATP
AMP
glycolysis
GHRL
resveratrol
caloric restriction
energy homeostasis
PRKAG1
PRKAG2
PRKAG3
gamma subunit
PRKAA1
T172
PRKAA2
alpha subunit
PRKAB1
PRKAB2
beta subunit
AMPK
cytosolic calcium ion homeostasis
Ca2+
CALM1
Ca2+
Calmodulin
CAMKK2
PRKAG1
PRKAG2
PRKAG3
gamma subunit
PRKAA1
PRKAA2
alpha subunit
PRKAB1
PRKAB2
beta subunit
AMPK
PRKN
UPS
GSK3B
DDIT4
DDIT4
P
T185
Y187
MAPK1
AKT1
S473
TSC2
TSC1
TSC1:TSC2
TSC2
TSC1
TSC1:TSC2
Akt pathway
mutation T415N
PRKN
mutation T240R
PRKN
mutation R42P
PRKN
PRKN
PHLPP1
mutated S18Y
UCHL1
UBE2N
UBE2V1
UBE2N:UBE2V1
MTOR
MLST8
RICTOR
MAPKAP1
PRR5
mTORC2
UCHL1
UCHL1
PRKN
UCHL1:PRKN
K63 Ub chain
UCHL1
RPS6KB1
T389
RPS6KB1
DEPDC5
NPRL2
NPRL3
GATOR1
epilepsy, familial focal
GATOR2
SESN2
MTOR
MLST8
RPTOR
DEPTOR
AKT1S1
mTORC1
MTOR
MLST8
RPTOR
DEPTOR
AKT1S1
mTORC1
RRAGA
RRAGC
RRAGB
RRAGD
RAGs
LAMTOR1
LAMTOR2
LAMTOR3
Ragulator
SLC38A9
Ragulator:SLC38A9
v-ATPase
GATOR2
SESN2
sestrin:GATOR2
leucine
SESN2
amino acids
amino acids
arginine
CUL4A
DDB1
DDB1:CUL4A
GTP
RHEB
GDP
RHEB
autophagy-lysosome system
lysosome
neuron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NPRL3
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GATOR2
SESN2
sestrin:GATOR2
SESN2
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